InstituteIQ: What Are We Building?

PS1 Institute Operations Copilot, explained in simple language

The problem in simple words

Training institutes (coaching classes) run on Excel sheets, WhatsApp groups and memory. Student details, fees, attendance and trainer timings are all in different places. The owner spends hours answering simple questions like "Who has not paid this month?" or "Which trainer is free on Wednesday?" and problems are noticed only after the damage is done.

Our solution: one portal with an AI assistant

InstituteIQ is a web portal that keeps everything in one place. On top of it sits an AI assistant called the Copilot. Instead of searching sheets, staff just type a question in normal English and the Copilot finds the answer.

Think of it as a smart manager who knows every student, fee, class and trainer, answers in seconds, but ALWAYS asks your permission before doing anything important.

What is inside the portal

Students and Batches
Add, search and manage students, courses, batches and trainers.

Fees and EMIs
See who paid, who is pending and whose EMI is overdue.

Attendance
Classroom check-in and online join, both recorded.

Timetable
Automatic clash-free weekly plan that re-plans in seconds.

AI Copilot
Ask questions in plain language and get answers with proof.

Approvals and Audit Log
Every change needs approval and is recorded forever.

Who will use it? How many users?

The portal is for institute STAFF. There are 4 types of users (called roles). Students do not log in; they are the records that staff manage. Each role sees only what it needs.

Role	In a real institute	In our demo	What they do
Owner	1 to 2	1 login	Sees everything, approves actions, controls the timetable
Counselor	2 to 3	1 login	Talks to students, checks batches and attendance, sends reminders
Trainer	5 to 10	1 login	Marks attendance, sees own batches. Can NEVER see fees
Accounts	1 to 2	1 login	Handles fees, EMIs and payments
Total	About 10 to 15 staff	4 logins	One demo login per role, so judges can switch roles

4
demo logins
(one per role)

~100
sample students
in the database

6
batches
(Java, React, Python, DSA...)

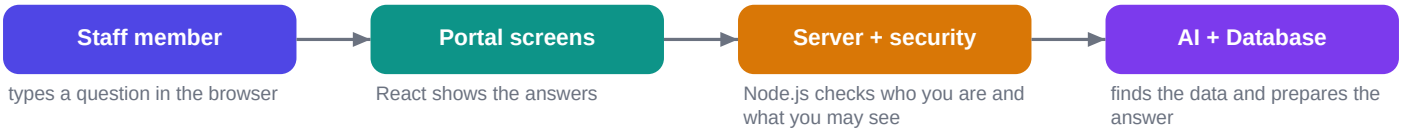
4
trainers and
3 rooms

These numbers are for the hackathon demo. The system works the same way for a bigger institute.

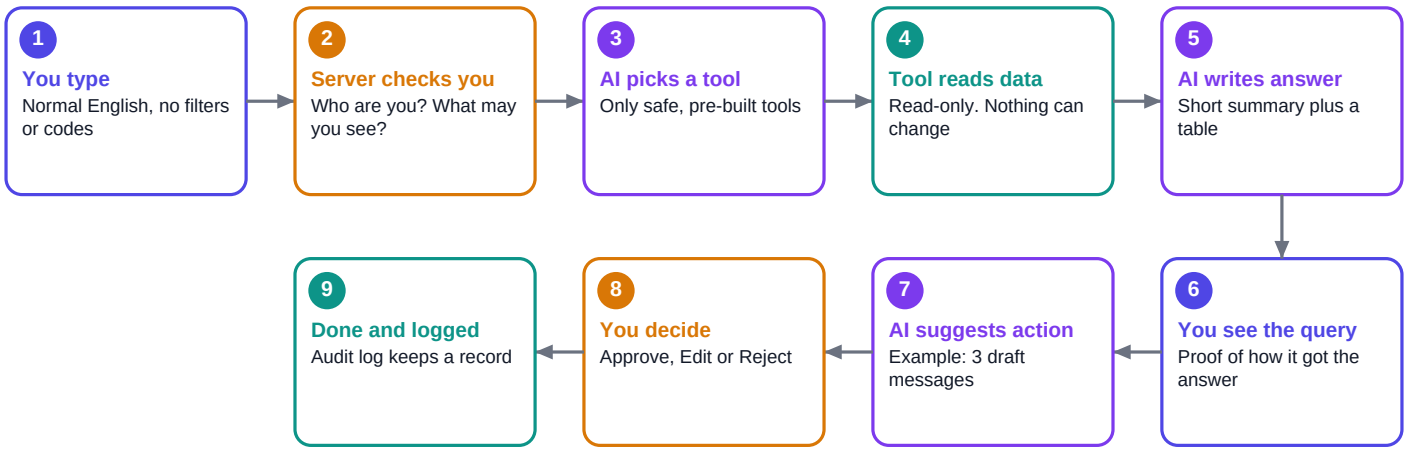
How It Works: The Complete Flow

From typing a question to an approved, recorded action

The big picture



Flow 1: Asking a question (the Copilot)



Example: a real conversation

You: Who has not paid their EMI this week?

Copilot: 3 students have an EMI due this week: Rahul Patil (Rs. 4,500, 18 Oct), Sneha Kulkarni (Rs. 4,500, 20 Oct), Omkar Jadhav (Rs. 6,000, 21 Oct). Query used: emis where status = PENDING and due date within 7 days. Want me to draft reminders?

You: Yes, draft reminders for them.

Copilot: I made 3 WhatsApp drafts. Nothing is sent yet. Please review.

Approve & Send | Edit | Reject

Flow 2: The timetable

Example: Trainer Amit is on leave on Wednesday. The system moves only the Java class to Thursday 2:00 PM (trainer and room are free) and explains why. The other 30 classes stay unchanged.

Why it is safe (4 simple rules)

- The AI can only READ data through pre-built safe tools. It can never change anything by itself.
- Anything that changes data or sends messages is shown as a preview first. A human must approve it.
- Roles are checked on the server. A trainer asking about fees gets a polite refusal, however the question is worded.
- Every question, query and approval is saved in the audit log with who, what and when.

What We Have To Do

Screens to build, the order of work and what must work in the final demo

Screens in the final project

Login | Dashboard | Students | Batches | Fees and EMIs | Attendance | Timetable | Copilot Chat | Approvals | Audit Log

Step-by-step work order

Step	What to do	What you can see when it is done
1	Set up React, Express and MongoDB in one repo	Blank app runs on your laptop
2	Create data models and a seed script	About 100 students and 6 batches in the database
3	Build login with 4 roles	Each role sees a different menu
4	Build the list screens (students, batches, fees, attendance)	The normal portal works, even without AI
5	Build the Copilot: LLM + 6 safe tools + chat screen	Ask a question and get an answer with the query
6	Add approvals and the audit log	Actions wait for approval and are recorded
7	Build the timetable and the re-plan button	Clash-free plan that updates on one click
8	Polish the screens and practise the demo	A smooth 3-minute demo

What must work in the final demo

- [ ] Owner asks the Java batch question and gets a table plus the query used
- [ ] Copilot drafts 3 WhatsApp reminders that wait for approval
- [ ] Approve button works and the action appears in the audit log
- [ ] Trainer asks about fees and is politely refused (and it is logged)
- [ ] Mark a trainer unavailable and the timetable re-plans with reasons

Simple words you will hear

Word	Simple meaning
Agent	An AI that plans steps, uses tools and checks results, instead of just answering once
Tool	A small ready-made function the AI may call, like "get absent students"
Role-based access	Each user type sees only the data it is allowed to see
EMI	A monthly instalment of a student's course fee
Query	The exact database search that produced an answer
Audit log	A permanent diary of who did what and when

Extras if time is left (bonus)

1) Dropout-risk flags with reasons 2) Match UPI or bank payments to EMIs 3) Completion certificates. For the full technical plan, see the 10-page blueprint PDF.